

Mustaqim Nurul Huda, S.Tr.T.

Technical Portfolio — Power Electronics, Embedded Systems & Automation

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Surabaya, Indonesia

Industrial Electrical Engineering graduate (B.A.Sc.) from Politeknik Elektronika Negeri Surabaya (PENS) with a GPA of 3.48/4.00. Specialized in hardware-firmware co-design, power converter topologies, intelligent MPPT algorithms, and industrial utility maintenance.

Technical Skill Matrix

Hardware: Schematic Capture, 2-Layer PCB Layout, High-Frequency Power Electronics (Modified Ćuk/SEPIC Converters), Isolated Sensing (AMC1100, ACS712), Optocoupler Gate Drivers (FOD3182), 3-Phase AC Motor Rewinding & Overhaul, Power Panel Assembly, Megger Insulation Diagnostics, BDV Oil Testing.

Software / Firmware: Embedded C/C++, Python (PyQt6 GUI, PyQtGraph, NumPy, Pandas, Matplotlib), MATLAB/Simulink, SQL (PostgreSQL), PLC Programming (Siemens TIA Portal, Omron CX-Programmer), SCADA Systems, Node-RED.

EDA & Engineering Tools: EAGLE PCB, Autodesk Fusion 360, AutoCAD Electrical, Autodesk Revit, Autodesk Inventor, ETAP, Solar-Log Telemetry, STM32CubeIDE, Git, VS Code.

Project Documentation

Project 1 — MPPT System Based on a Hybrid WODE Algorithm

Undergraduate Thesis, Politeknik Elektronika Negeri Surabaya

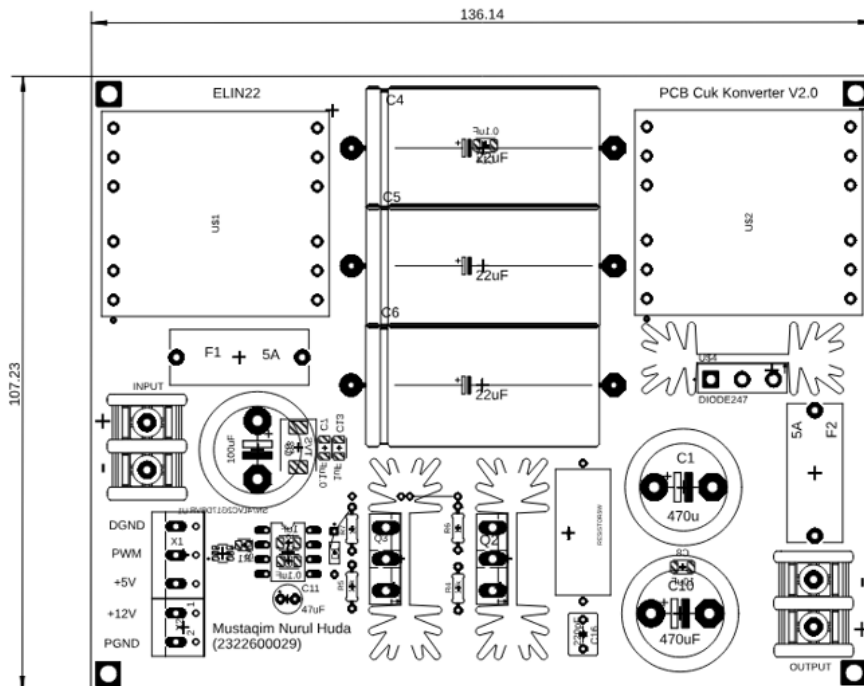
Feb 2024 – Jul 2026

1. The Problem

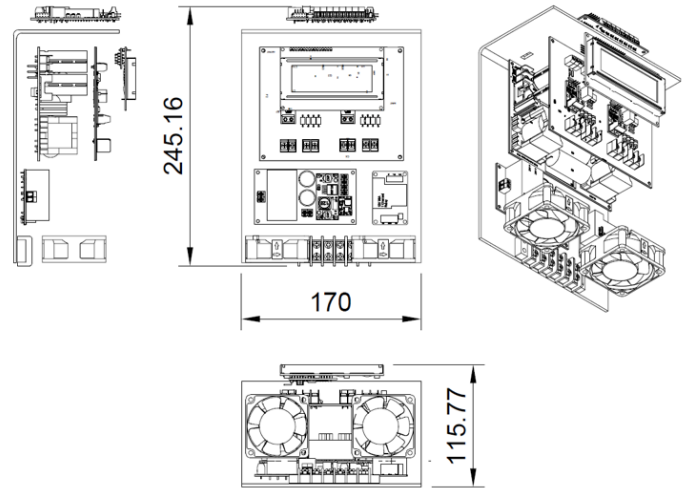
Standalone solar pumping systems suffer severe power degradation and pump stalling under Partial Shading Conditions (PSC) due to multiple local power peaks. Conventional MPPT algorithms get trapped in local maxima, causing erratic hydraulic discharge.

2. Schematics & PCB Layouts

Engineered a custom 2-layer PCB for a 50 kHz Modified Ćuk DC-DC converter using EAGLE. Integrated isolated AMC1100 feedback amplifiers, ACS712 Hall-effect current sensors, and FOD3182 optocoupler gate drivers. Modeled 3D enclosures in Autodesk Fusion 360 alongside production 2D technical drawings.



Modified Ćuk Converter PCB Design V2.0



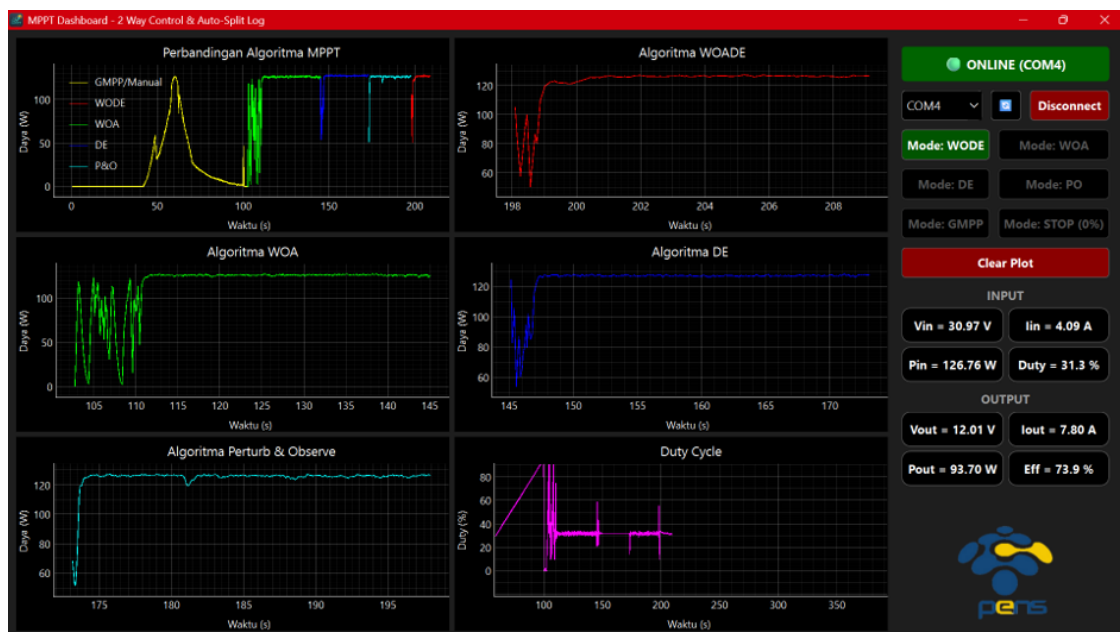
Technical Drawing of Controller Housing

3D Housing / Enclosure Design of MPPT Controller

3. Firmware & Software Architecture

Programmed an ARM Cortex-M4 STM32F407 microcontroller in C++ executing a hybrid WODE algorithm with dynamic slew-rate limiting. Streamed serial telemetry via UART/Bluetooth to a custom Python PyQt6 desktop GUI for real-time tracking, parameter tuning, and high-frequency data logging.

Public Repository: github.com/Prospexxx



GUI Dashboard (PyQt6) — Real-Time Monitoring & MPPT Algorithm Comparison

4. Testing & Validation

Validated hardware performance over 86,790 experimental data samples across 5 irradiance patterns. Achieved an average tracking efficiency of **98.53%**, convergence speed of **3.67 seconds**, and steady-state power ripple of **2.28 W**. Research submitted to *Indonesian Journal of Science & Technology* (IJOST) and patent filed with DJKI.

5. Your Specific Role

Sole Lead Hardware & Firmware Designer. Responsible for converter topology selection, PCB layout, C++ algorithm implementation, Python PyQt6 software development, and full field validation.

Project 2 — EcoMow: Battery-Powered Electric Lawn Mower

Team Lead, PKM-KC Research Grantee

May 2024 – Nov 2024

1. The Problem

Conventional combustion-engine lawn mowers generate high noise pollution, carbon emissions, and operating fuel costs in institutional landscaping. A zero-emission electric alternative was required to deliver optimal cutting torque while maintaining battery operational endurance.

2. Schematics & PCB Layouts

Drafted high-current power distribution schematics, protection circuitry, and BMS wiring layouts. Modeled the 3D mechanical frame, cutting deck, and protective outer housing using Autodesk Fusion 360.

3. Firmware & Software Architecture

Implemented motor control interlock logic and overcurrent protection routines. Authored official technical manuals and structural guidebooks submitted for national intellectual property protection.

4. Testing & Validation

Constructed a fully functional zero-emission prototype funded by the national PKM-KC grant. Passed field operational tests with significant noise suppression and zero direct emissions. Secured Copyright Registration (HAKI No. EC00202499819).



PKM-KC Grant Award Certificate (No. 449/D4/DV.01.05/2024)



Copyright Registration Certificate (HAKI No. EC00202499819)

5. Your Specific Role

Project Lead & Primary Hardware Engineer. Directed a 4-member multidisciplinary engineering team, managed project timelines and budgets, designed propulsion circuitry, and secured copyright registration.

Project 3 — Performance Diagnostics & Utility Maintenance (1.37 MWp Solar Plant)

Electrical Engineering Intern, PT Eratex Djaja Tbk

Mar 2025 – Jun 2025

1. The Problem

Industrial facilities face major financial penalties if power factor degrades or if high-capacity solar assets underperform. Unscheduled downtime in 1.37 MWp PV systems or power transformers disrupts textile plant operations.

2. Schematics & PCB Layouts

Reviewed Single Line Diagrams (SLD) and control panel schematics for Sungrow SG110CX inverters and Canadian Solar arrays. Planned layout, assembled, and direct-wired 3-phase power distribution and control panels (MCCBs, MCBs, contactors).

3. Firmware & Software Architecture

Utilized Solar-Log PV telemetry software for real-time energy yield auditing, specific performance evaluation, and power factor monitoring.

4. Testing & Validation

Executed Megger insulation resistance testing, PV string I-V curve tracing, transformer oil vacuum dehydration purification with Dielectric Breakdown Voltage (BDV) tests, and 3-phase AC motor overhauls under strict LOTO safety protocols. Confirmed stable power factor ($\cos \varphi = 0.967$) and solar energy coverage of **13.6%** of factory demand.



Internship Certificate — PT Eratex Djaja Tbk

5. Your Specific Role

Electrical Engineering Intern. Conducted hands-on panel assembly, high-voltage diagnostic testing, transformer oil purification, motor coil rewinding, and Solar-Log performance reporting.

Publications, Patents & Academic Credentials

Simple Patent (DJKI): Standalone WODE-MPPT Solar Controller for DC Submersible Pumps (2026).

Copyrights (HAKI Kemenkumham RI): Registered Copyrights for EcoMow Technical Manual (EC00202499819) and Real-Time MPPT Python PyQt6 Dashboard Software (2024–2026).

Language Proficiency: PENS English Proficiency Test (PEPT) Score: **527** (Equivalent to TOEFL ITP).

Organizational Leadership: Active Member at HIMA ELIN (2022–2026) and UKM Dirgantara PENS (2023–2025).